Recursion

Programming for Problem Solving (PPS)
GTU # 3110003

USING

{C}
Programming

Prof. Nilesh Gambhava

Computer Engineering Department,
Darshan Institute of Engineering & Technology, Rajkot

What is Recursion?

- Any function which calls itself is called recursive function and such function calls are called recursive calls.
- Recursion cannot be applied to all problems, but it is more useful for the tasks that can be defined in terms of a similar subtask.
- It is idea of representing problem a with smaller problems.
- Any problem that can be solved recursively can be solved iteratively.
- When recursive function call itself, the memory for called function allocated and different copy of the local variable is created for each function call.
- Some of the problem best suitable for recursion are
  - Factorial
  - Fibonacci
  - Tower of Hanoi

Working of Recursive function

Working

```
void func1();


void main()
{
  ....
  func1();
  ....
}

void func1()
{
  ....
  func1();
  ....
}
```

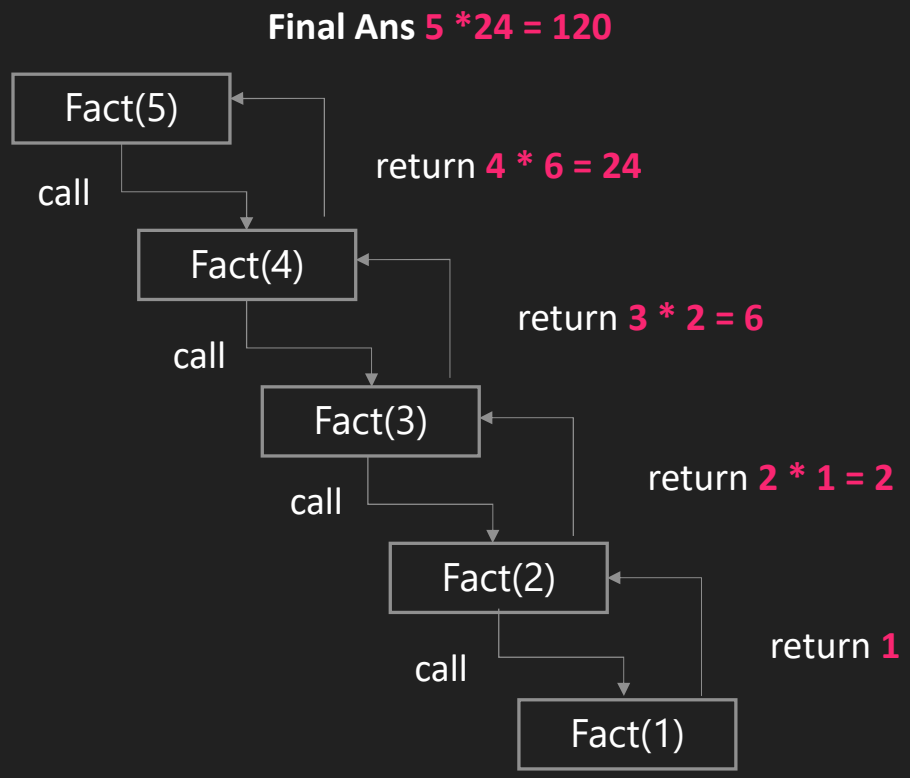
Properties of Recursion

- A recursive function can go infinite like a loop. To avoid infinite running of recursive function, there are two properties that a recursive function must have.
- Base Case or Base criteria
  - It allows the recursion algorithm to stop.
  - A base case is typically a problem that is small enough to solve directly.
- Progressive approach
  - A recursive algorithm must change its state in such a way that it moves forward to the base case.

Recursion - factorial example

- The factorial of a integer n, is product of
  - n * (n-1) * (n-2) * …. * 1
- Recursive definition of factorial
  - n! = n * (n-1)!
  - Example
    - 3! = 3 * 2 * 1
    - 3! = 3 * (2 * 1)
    - 3! = 3 * (2!)

Recursive trace



WAP to find factorial of given number using Recursion

Program

```c
#include <stdio.h>
int fact(int);
void main()
{
    int n, f;
    printf("Enter the number?\n");
    scanf("%d", &n);
    f = fact(n);
    printf("factorial = %d", f);
}
int fact(int n)
{
    if (n == 0)
        return 1;
    else if (n == 1)
        return 1;
    else
        return n * fact(n - 1);
}
```

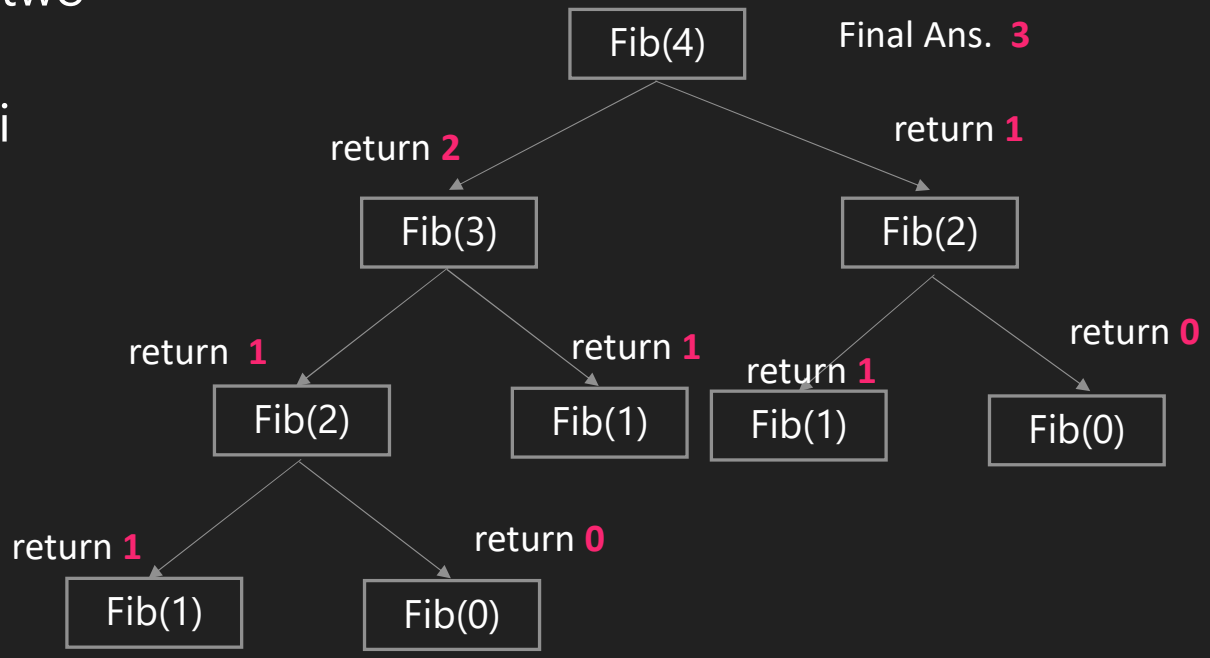
Output

```
Enter the number? 5
factorial = 120
```

Recursion - Fibonacci example

- A series of numbers , where next number is found by adding the two number before it.
- Recursive definition of Fibonacci
  - Fib(0) = 0
  - Fib(1) = 1
  - Fib(n) = Fib(n-1) + Fib(n-2)
- Example
  - Fib(4) = Fib(3) + Fib(2)
  - Fib(4) = 3

Recursive trace



WAP to Display Fibonacci Sequence

Program

```c
#include <stdio.h>
int fibonacci(int);
void main()
{
    int n, m = 0, i;
    printf("Enter Total terms\n");
    scanf("%d", &n);
    printf("Fibonacci series\n");
    for (i = 1; i <= n; i++)
    {
        printf("%d ", fibonacci(m));
        m++;
    }
}
```

Program contd.

```c
int fibonacci(int n)
{
    if (n == 0 || n == 1)
        return n;
    else
        return (fibonacci(n - 1) +
    fibonacci(n - 2));
}
```

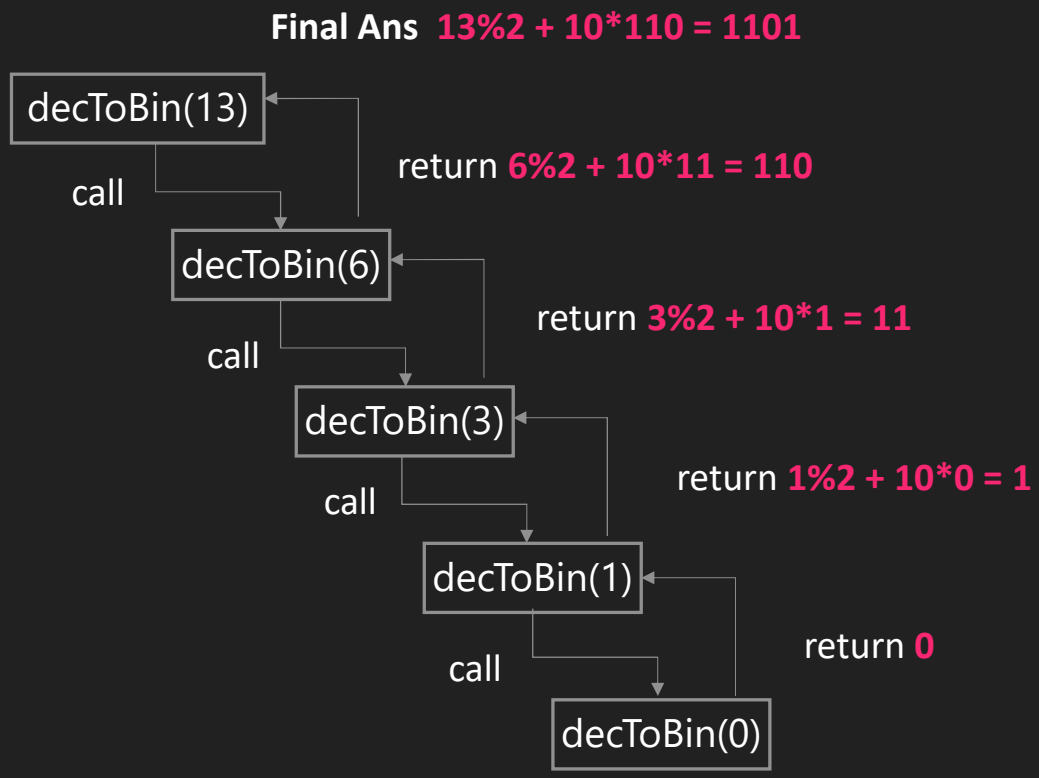
Output

```
Enter Total terms
5
Fibonacci series
0 1 1 2 3
```

Recursion - Decimal to Binary example

- To convert decimal to binary, divide decimal number by 2 till dividend will be less then 2
- To convert decimal 13 to binary
  - 13/2 = 6 reminder 1
  - 6/2 = 6 reminder 0
  - 3/2 = 3 reminder 1
  - 1/2 = 1 reminder 1
- Recursive definition of Decimal to Binary
  - decToBin(0) = 0
  - decToBin(n) = n%2 + 10* decToBin(n/2)
- Example
  - decToBin(13) = 13%2 + 10 decToBin(6)
  - decToBin(13) = 1101

Recursive trace



WAP to Convert Decimal to Binary

Program

```c
#include <stdio.h>
int convertDecimalToBinary(int);
void main()
{
    int dec, bin;
    printf("Enter a decimal number: ");
    scanf("%d", &dec);
    bin = convertDecimalToBinary(dec);
    printf("The binary equivalent = %d \n",bin);
}
int convertDecimalToBinary(int dec)
{
    if (dec == 0)
        return 0;
    else
        return (dec % 2 + 10 *
    convertDecimalToBinary(dec / 2));
}
```

Output

```
Enter a decimal number: 12
The binary equivalent = 1100
```

WAP to Convert Binary to Decimal

Program

```c
#include <stdio.h>
int convertBinaryToDecimal(int b, int c, int t);
void main()
{
    unsigned int binary, decimal;
    printf("Enter a binary number: ");
    scanf("%d", &binary);
    decimal = convertBinaryToDecimal(binary, 1, 0);
    printf("Decimal value of %d is %d", binary, decimal);
}
int convertBinaryToDecimal(int b, int c, int t)
{
    if (b > 0)
    {
        t += (b % 10) * c;
        convertBinaryToDecimal(b / 10, c * 2, t);
    }
    else
        return t;
}
```

Output

```
Enter a binary number: 101
Decimal value of 101 is 5
```

Practice Programs

1) Write a program to find factorial of a given number using recursion.
2) WAP to convert decimal number into binary using recursion.
3) WAP to use recursive calls to evaluate $F(x) = x - x^3/3! + x^5/5! - x^7/7! + \ldots + x^n/n!$

Thank you